

Choosing The Right SOUND LEVEL METER



Biswajit Das is manager R&D, EFY Labs

When performing acoustic testing, choosing a proper sound level meter (SLM) testing system is the key. An SLM is commonly used in noise pollution studies for quantification of different kinds of noise, especially industrial, environmental and aircraft. However, reading from an SLM does not correlate well with human-perceived loudness, which is better measured by a loudness meter.

Types of SLMs

Integrating meter. An integrating meter is one that can measure an equivalent continuous sound level (Leq) and is required for any proper occupational or environmental noise monitoring. This meter is usually mandated.

Non-integrating meter. A non-integrating meter displays only the instantaneous noise level at any one time and is suitable only for spot checks and very steady noise levels.

As per IEC standards, SLMs of the two classes, Class 1 and Class 2, have the same functionality but different tolerances for errors. Class 1 instruments have a wider

frequency range and a tighter tolerance than lower-cost, Class 2 units.

ANSI specifies SLM as three different types, namely, 0, 1 and 2. Type 0 is used in laboratories, type 1 for precision measurements in the field and type 2 for general-purpose measurements. Type 1 meter is preferred for designing cost-effective noise controls.

Frequency weighting

Frequency weighting selections allow users to choose how the meter will treat sound measurements over its specified frequency range. Certified SLMs offer noise measurements with A, C and Z frequency weighting.

A human ear responds more to frequencies between 500Hz and 8kHz and is less sensitive to very-low-pitch or high-pitch noises. In A weighting (expressed as dBA), the meter gives more weight to certain frequencies. In other words, sound levels at certain frequencies are boosted or cut (to match how the human ear responds to sound).

In C weighting (expressed as dBC), all frequencies are treated the same (no boosting or cutting of sound level).

Z weighting (expressed as dBZ) is a flat frequency response of 10Hz to 20kHz ± 1.5 dB.

Frequency weighting A is the only weighting that is mandated by the international standard; frequency weightings C and Z are optional fits.

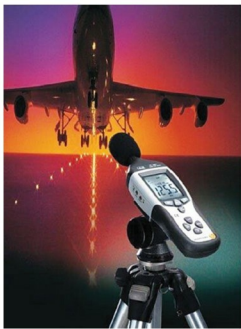


Fig. 1: A sound level meter in use

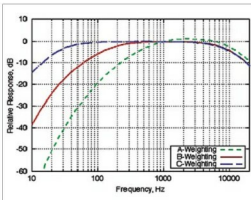


Fig. 2: A, C and Z frequency weightings for sound